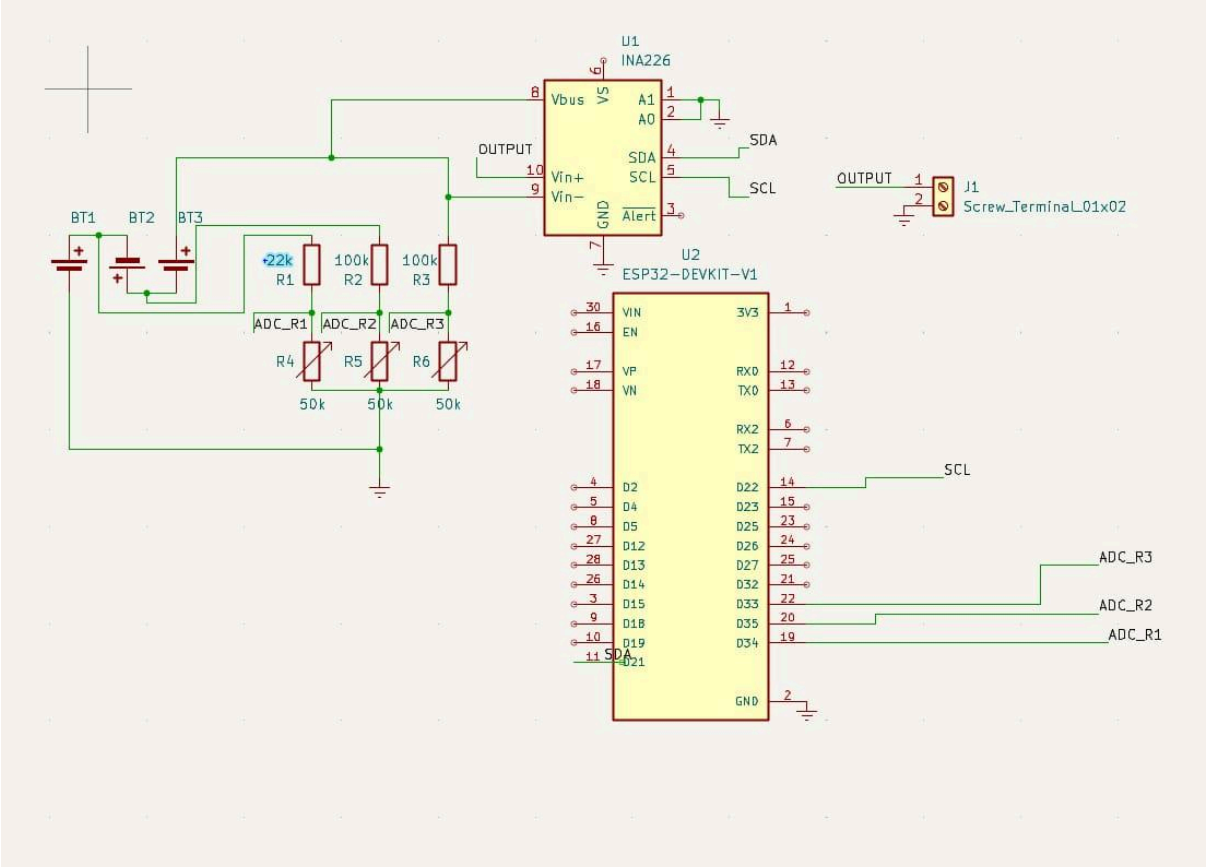
CURRENT STATUS

1. Collected and reviewed literature on battery monitoring, SOC, SOH and RUL estimation techniques.

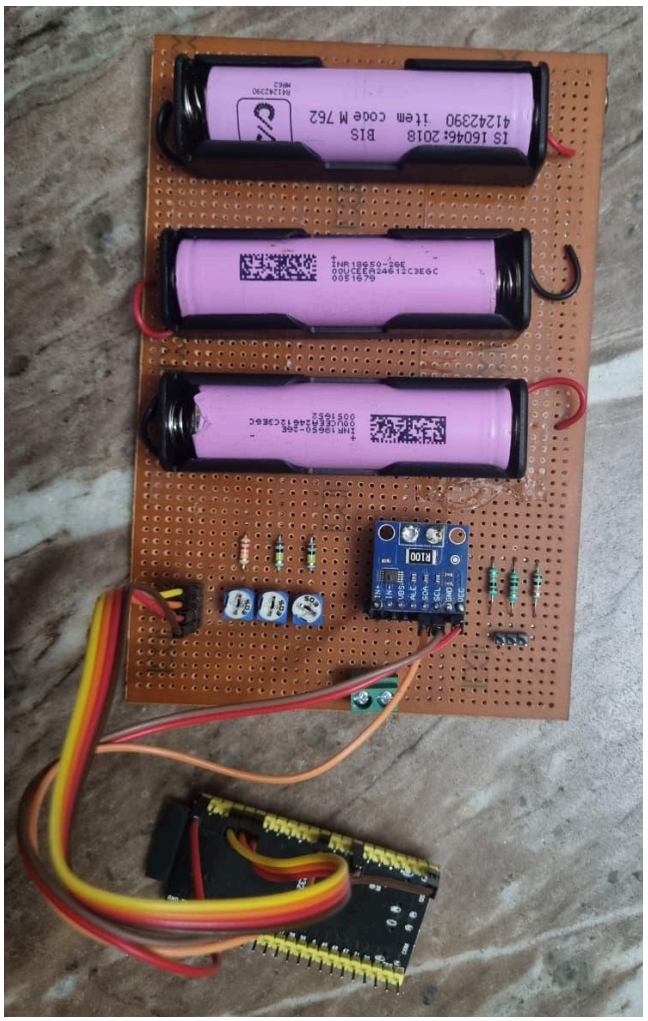
2. Selected the required hardware and software resources.

3. Developed the initial Flutter dashboard for battery monitoring.

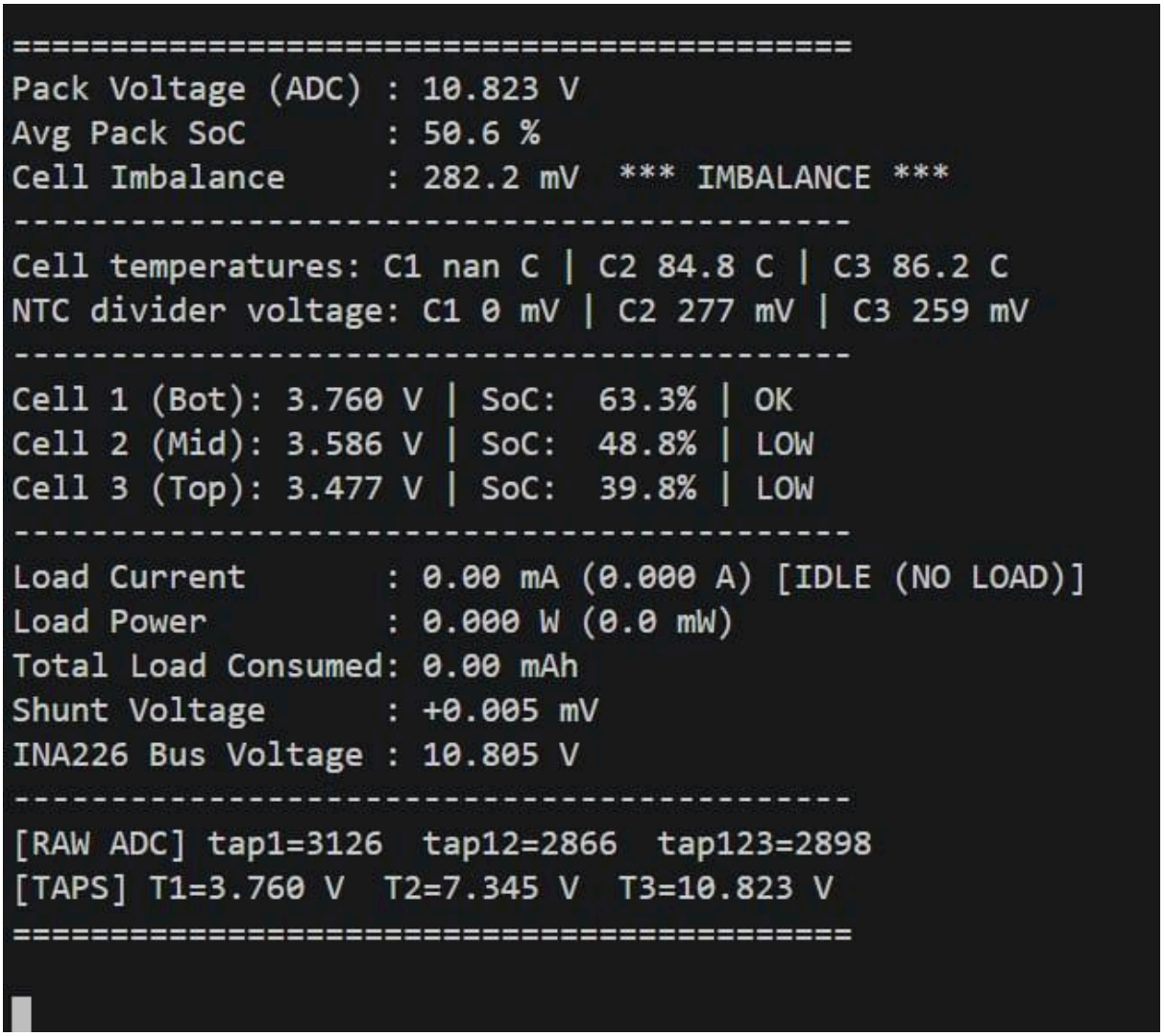
4.Developed the initial hardware setup for voltage and current measurement; temperature sensing is yet to be added.



Circuit Diagram of the Initial Hardware Setup



Hardware Setup for Voltage and Current Measurement

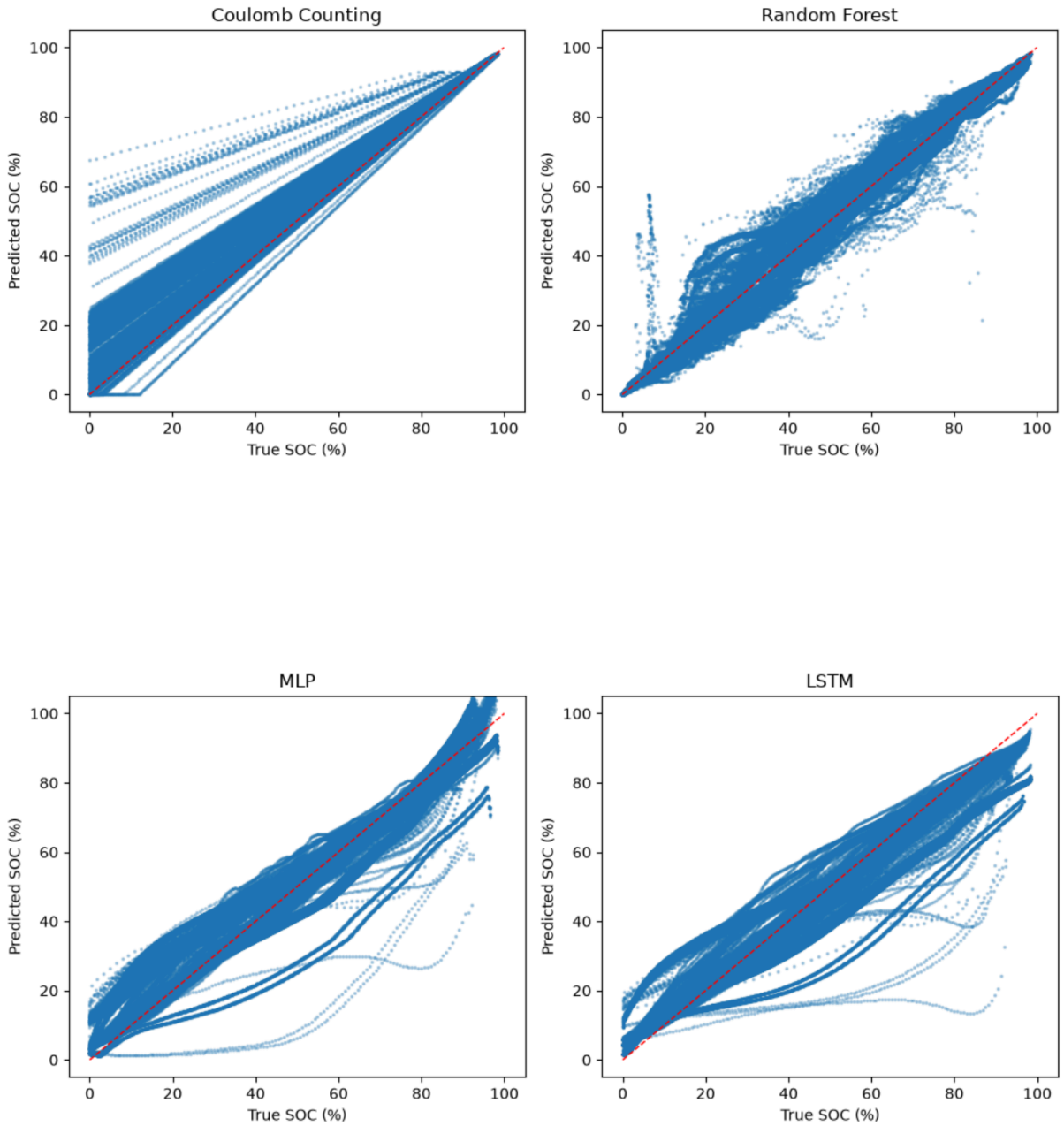


VS Code Terminal Output of Hardware Setup

5. Collected the NASA Li-ion battery dataset, performed exploratory data analysis and preprocessing.

6. Developed SOC estimation models using Coulomb Counting, Random Forest, MLP, and LSTM.

7. Compared the developed models using error and performance metrics.



Comparison of SOC Estimation Models

Model	MAE (%)	RMSE (%)	R2
Coulomb Counting	4.27	7.54	0.927
Random Forest	3.59	5.15	**0.966**
MLP	6.19	7.89	0.921
LSTM	6.70	8.68	0.904

Performance Metrics of SOC Estimation Models